



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. CSE – B – Section

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Dr.R.Venkatesh

Innovative Practice Description

Unit / Topic: Unit III / **Iteration:** state, while, for

Course Outcome: CO 3

Topic Learning Outcome: TLO 8

Activity Chosen: Think Pair Share

Objective:

Think-Pair-Share (TPS), which involves students in a systematic process of individual reflection, peer conversation, and group sharing, aims to foster collaborative problem-solving abilities and active learning. To encourage cooperative learning and problem-solving abilities via interaction and exchanged insights in order to gain deeper understanding of iterative constructions (while and for loops) and their use in programming.

Time Allotted for the Activity: 15 minutes

Procedure of the Implementation:

- Collaborative, active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).

Preparation Phase

- Clearly define the activity's learning objectives, making sure they relate to the subject or idea being taught.
- Create problems, scenarios, or open-ended questions that promote conversation and critical thinking.
For example: "How can you optimize a for loop to improve performance in large datasets?"
- Before starting, provided students the background information or instructions they needed on the topic at hand.

T (Think): Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 1-5 Minutes].

P (Pair): Teacher asks another question, related to the previous one, that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 2-5 Minutes].

S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 05-10 minutes].

Details of the Implementation

Roles assigned to the Students

Assigning specific roles to students in a Think-Pair-Share activity helps ensure structured participation, clear responsibilities, and effective collaboration.

- **Each student** reflects on the problem or question individually, formulates their initial thoughts, and writes down their response.
- **One of student explains** their individual solution or thought process to **their partner**.
- **Reporter: Breetta J and Rhema Shinny K act as reporters.** They are responsible for summarizing the pair's discussion and presenting it to the larger group or class.

1. Initial Teaching and Material Distribution

- The teacher begins by teaching the topics related to iteration (while, for, and state) using engaging examples and relevant concepts. Supporting materials, such as lecture notes, code snippets, and real-world applications, are provided to the students for reference and understanding.
- The teacher presents a set of strategy-focused or real-world business-related questions based on the topic.
Example: *"Design a loop structure to calculate employee overtime wages dynamically for an organization with varying shift durations."*
- Students are asked to form groups and given 5 minutes to individually think about solutions to the given questions.

Individual Reflection:

- During this phase, students individually analyze the problem and come up with their own answers and solutions.
- This step ensures that every student contributes original ideas to the activity. It shows in Figure 1.

Pair Sharing:

- Students share their answers and ideas within their groups.
- They compare their solutions, discuss various approaches, and collaborate to refine their understanding and answers. It shows in Figure 1.

Class-Wide Sharing:

- **Figure 2** shows **Breetta J and Rhema Shinny K** present their solutions and reasoning to the entire class.
- The teacher facilitates a discussion, encouraging questions and feedback from other students.

Discussion and Consolidation:

- The teacher leads a final discussion summarizing all answers and providing feedback on their feasibility, correctness, and efficiency.
- Misconceptions are addressed, and additional insights are provided to connect the activity with broader real-world applications.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO3	PSO1
CO 3	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	2	1	1	1
Justification for correlation	Students can be able to apply logical thinking and think about problem-solving approaches.	Students will be able to approach the problem by applying the types of operators they learned and solving simple and complex engineering problems.	Students will be able to interpret the complex engineering problems using the precedence of operators.	The problem solving skill earned through this activity helps the students in solving various problems in information and communication engineering.

Images / Screenshot of the practice:



Figure:1 Individual Reflection and Pair Sharing



Figure:2 Breetta J and Rhema Shinny K present their solutions

Reflective Critique:***Feedback of practice from students and other stakeholders:***

- The chance to explore a variety of topics and participate in group learning was valued by the students.
- They discovered that the business-based, real-world questions were enlightening and useful, assisting them in comprehending how the ideas relate outside of the classroom.
- Many emphasized how the structured phases—think, pair, and share—helped them become more confident and develop their critical thinking abilities.
- In order to develop thorough answers, some students asked for additional time during the "think" portion.

Benefit of the practice:

- Encourages students to deeply reflect on the topic, improving their conceptual clarity.
- Discussing with peers helps uncover alternative approaches and clears misconceptions.
- Promotes analytical thinking by requiring students to evaluate and justify their answers.
- Enables students to approach problems from multiple perspectives.
- Fosters teamwork through pair and group discussions.
- Enhances articulation skills as students explain their ideas to peers and the class.
- Transforms a passive learning environment into an interactive and participatory one.

Challenges faced in implementation:

- Allocating sufficient time for each phase (think, pair, and share) without exceeding class duration can be challenging.
- Some groups may require more time to discuss or present, causing delays.
- Dominant students may overshadow quieter ones, leading to unequal contributions.
- Some students might feel reluctant to participate due to shyness or lack of confidence.
- Students with weaker foundational knowledge might struggle to contribute meaningfully, especially in the "think" phase.
- Advanced students may feel the activity is too basic or repetitive.
- Pair or group conflicts can hinder effective collaboration.
- Mismatched abilities within pairs/groups can result in one-sided participation.

References:

1. adlit.org, “Adolescent Literacy”, 2019. [Online]. Available: <http://www.adlit.org/strategies/23277/> [Accessed: 23.01.2020]
2. Reading Books - <https://www.readingrockets.org/strategies/think-pair-share>
3. prodigygame.com, “active-learning-strategies”, 2019. [Online]. Available: <https://www.prodigygame.com/blog/active-learning-strategies-examples/>
4. Mr.R.Venkatesh., “Introduction to Yoga and meditation for professional excellence and stress management.” in course GE6075 Professional Ethics in engineering <https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf>

Signature of Faculty Member**HOD-CSE**